Converted vs. Unconverted Photons: Shower Shape, Isolation, and Energy Response in photon10 MC

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Abstract

We study the impact of photon conversions (pre-EMCal e^+e^- pair production in detector material) on reconstruction observables used by the PPG12 isolated-photon cross-section measurement in p+p at $\sqrt{s} = 200$ GeV. Using the `photon10` single-particle Monte Carlo ($\sim 10\,\mathrm{M}$ events, truth $p_T \in [14, 30]$ GeV), we find that **17.3%** of fiducial truth photons ($|\eta| < 0.7$, $14 \leq p_T < 30$ GeV) convert before the EMCal. Converted photons exhibit (i) a mean cluster-energy response shift of $\Delta\langle R_E\rangle \simeq -0.076$ (-8%) driven by an asymmetric low-R tail, (ii) a drop of -14.6 percentage points in the parametric reco-isolation pass fraction (rising to -22.8 pp at 8–10 GeV), dominated by additional EMCal energy inside the isolation cone, and (iii) a -16% shift in the mean shower-shape BDT score, with w_ϕ^{CoG} broadening by $+91\%$ due to the solenoidal-field opening of the e^+e^- pair. These four features are mutually consistent with a single physical picture: the ~ 1.4 T sPHENIX solenoid splits the e^+e^- pair in azimuth, one arm leaks outside the cluster footprint (reducing the energy response and polluting the isolation cone), and the resulting wider, two-lobed ϕ-profile degrades the ID BDT. The effect enters the cross-section measurement as a joint ID+iso efficiency systematic whose size is set by the accuracy of the simulated material budget.

1 Introduction and Motivation

Photon conversion — the reaction $\gamma \to e^+e^-$ induced by the Coulomb field of a nucleus — occurs with probability $P \simeq 1 - \exp\left[-\frac{7}{9}\,X/X_0\right]$ as a photon traverses a thickness X/X_0 of material. In sPHENIX the tracker (MVTX + INTT + TPC) together with its support structures and cables precedes the barrel EMCal. With the nominally quoted $\sim$ 10–20% X_0 of material, a $\sim$ 10–15% conversion probability is expected in the central barrel. Once a photon converts, the two leptons are bent in opposite azimuthal directions by the ~ 1.4 T solenoid, land in the EMCal as two partially overlapping electromagnetic showers, and are reconstructed by the standard clustering as a single (wider) cluster whose properties differ from those of a pristine photon.

These distortions matter for PPG12 because the tight-selected isolated photon signal efficiency is the product of a shower-shape identification efficiency (via the E_T-dependent BDT cut) and an isolation efficiency (via the parametric isolation cone cut). Both steps are sensitive to the converted fraction and its pre-reconstruction kinematics. Any mismatch in the simulated material budget between Monte Carlo and data therefore propagates directly into the cross-section. Quantifying the conversion fingerprint on each observable — and its p_T and η dependence — is a prerequisite for assigning a material-budget systematic uncertainty.

Truth conversion information is carried in the slimtree through the `particle_converted` flag written by `anatreemaker/source/CaloAna24.cc`. The flag is built by traversing the G4 shower particle list of each truth photon (`CaloAna24.cc` lines $\sim$ 807–843): any secondary that (a) originates at a G4 vertex with $r_\perp < 93$ cm (i.e. upstream of the EMCal front face) and (b) carries more than 40% of the photon momentum, tags the photon. If at least one such secondary is an $e^\pm$ ($|\mathrm{pid}| = 11$), the photon is labelled `converted`; otherwise it is labelled `bad`. The flag is written at `CaloAna24.cc` lines $\sim$ 953–970 using:

$$\texttt{particle\_converted} = \begin{cases} 0 & \text{no upstream secondary above threshold (unconverted)} \\ 1 & e^+e^- \text{ daughter above threshold (converted)} \\ 2 & \text{non-}e^+e^- \text{ secondary above threshold (bad)} \end{cases} \tag{1}$$

In what follows we treat the small "bad" population (0.01–0.02% of truth photons, < 300 matched in 10 M events) as physically negligible and focus on the unconverted–converted comparison.

2 Data Sample and Method

Sample. The study is based on the single-photon Monte Carlo file `photon10_with_bdt_vtx_noreweight_single.root` (one fired photon per event, truth $p_T \in [14, 30]$ GeV at generation, $|\eta| < 1.5$). Four independent analyses read the tree `slimtree`:

- Fractions, shower shape, and energy response use the full 9,998,541 events.
- Isolation uses a 2,000,000-event subsample (runtime-based).

Truth selection (fiducial). `particle_pid` == 22, |`particle_Eta`| < 0.7, `particle_truth_iso_03` < 4 GeV, and (where applicable) 8 ≤ `particle_Pt` ≤ 36 GeV or 14 ≤ `particle_Pt` < 30 GeV as indicated.

Cluster–truth matching. For each truth photon the highest-E_T cluster in `CLUSTERINFO_CEMC` whose `cluster_truthtrkID` equals the truth `particle_trkid` is selected. A residual cluster E_T threshold (> 3–5 GeV, depending on the analysis) is applied. pT binning follows the PPG12 analysis edges `ptRanges` = $\{8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 32, 36\}$ (12 bins) with the matched cluster E_T as the abscissa.

Caveat on isolation. The input file *does not* contain the topo-cluster isolation branches `cluster_iso_topo_03/04`. The isolation study therefore uses the calo-tower isolation `cluster_iso_03` (and R=0.4 equivalent). The nominal analysis configuration `efficiencytool/config_bdt_nom.yaml` uses topo-iso at $R = 0.4$ (`use_topo_iso: 2`). The *shape* of the converted/unconverted difference is expected to carry over, but absolute magnitudes should be re-measured on topo-iso branches once they are populated for `photon10`.

Shower shape caveat. The shower-shape BDT score is only written to the `CLUSTERINFO_CEMC_NO_SPLIT` node, whereas the other shower-shape observables use `CLUSTERINFO_CEMC`. BDT conclusions below are therefore based on the unsplit-cluster node.

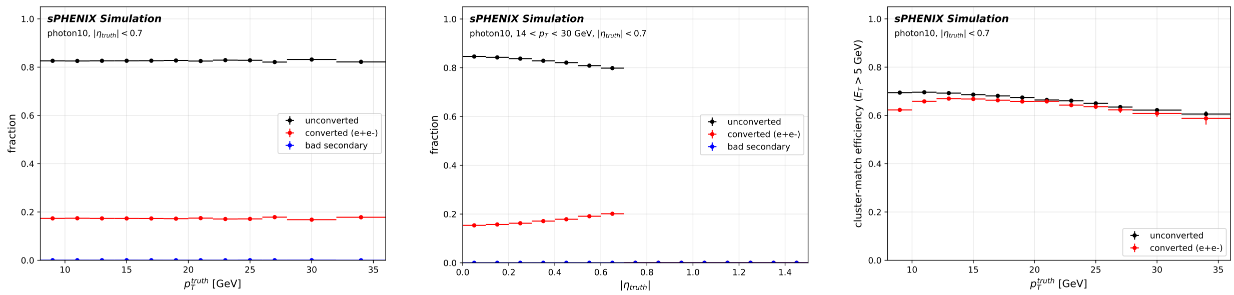
3 Conversion Fractions and Cluster Match Efficiency

In the fiducial region ($|\eta| < 0.7$, $14 \leq p_T < 30$ GeV, 855,380 truth photons), the converted fraction is $(17.317 \pm 0.041)\%$ and the "bad" fraction is a negligible $(0.024 \pm 0.002)\%$. Table 1 summarises the inclusive category counts and per-category cluster-match efficiency. The converted fraction is effectively flat in p_T (17.2–17.9% across 8–36 GeV) and rises with $|\eta|$, from $(15.4\pm0.3)\%$ at $|\eta| < 0.1$ to $(20.1 \pm 0.4)\%$ at $0.6 < |\eta| < 0.7$, reaching 32% at $|\eta| \simeq 1.4$–1.5 (extended window).

Table 1: Converted-photon fractions and cluster-match efficiency in the fiducial window $|\eta| < 0.7$, $14 \le p_T < 30$ GeV. Match efficiency is defined as the fraction of truth photons (within a category) for which at least one cluster satisfies `cluster_truthtrkID == particle_trkid` and `cluster_Et` > 5 GeV.

Category	N (truth)	Fraction	Match efficiency
Unconverted	707,047	$(82.659 \pm 0.041)\%$	$(67.987 \pm 0.055)\%$
Converted (e^+e^-)	148,129	$(17.317 \pm 0.041)\%$	$(66.290 \pm 0.123)\%$
Bad secondary	204	$(0.024 \pm 0.002)\%$	$(39.024 \pm 3.407)\%$

The three panels of Fig. 1 show the main trends:



(a) Fractions vs. p_T ($|\eta| < 0.7$)

(b) Fractions vs. $|\eta|$ ($14 \le p_T < 30$ GeV)

(c) Cluster-match efficiency vs. p_T

Figure 1: Converted-photon fractions and match efficiency in the fiducial window. The converted fraction is flat in p_T and rises with $|\eta|$ consistent with increased integrated material at forward rapidity. Converted photons have ~ 2 pp lower cluster-match efficiency than unconverted.

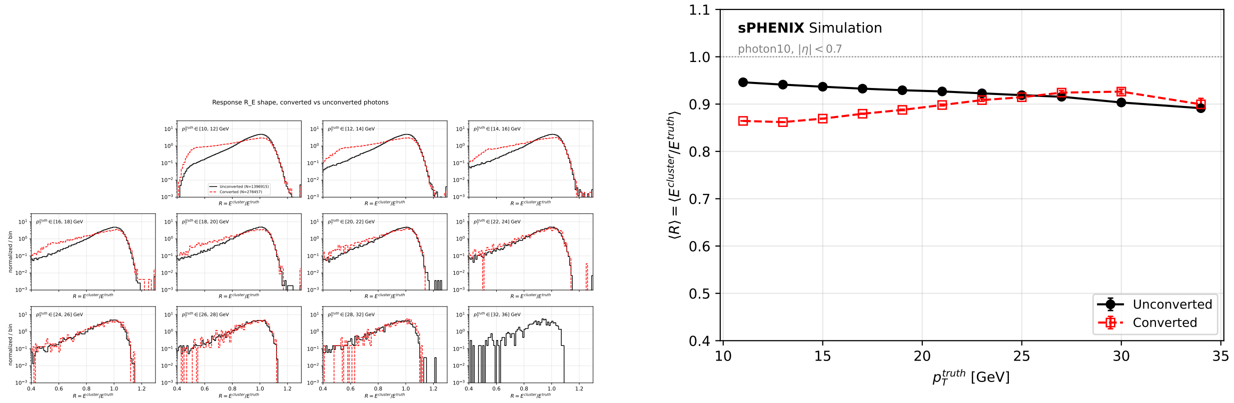
The central-barrel converted fraction of $\sim 17\%$ is consistent with a material budget near the upper end of the nominally quoted 10–20% X_0 range ($1 - \exp[-\frac{7}{9} \cdot 0.20] \simeq 15\%$). The sharp rise toward $|\eta| \sim 1.5$ is consistent with endcap-style material growth and matches expectations for the tracker support geometry.

The absolute match-efficiency value ($\sim$ 66–68% for fiducial photons) looks low at first glance because the definition *requires* `cluster_truthtrkID` to equal the *primary* photon track id. When clustering associates a post-shower secondary ($e^{\pm}$, brem γ) instead of the original photon, the match fails even though a physically correct cluster is present. The converted-category match efficiency is therefore a *lower bound*; the *relative* $\sim 2.5\%$ deficit with respect to unconverted is the physics-meaningful quantity.

4 Energy Response and Resolution

We define the cluster energy response $R_E = E_{\mathrm{cluster}}/E_{\gamma}^{\mathrm{truth}}$ and compare its shape, first and second moments, core Gaussian and Crystal Ball fit parameters, and low-R tail fractions between the converted and unconverted categories. Analogous results for R_{E_T} are contained in the auxiliary grid `response_shape_ET_grid.pdf`. The abscissa is the *truth* photon p_T (so that the per-bin efficiency reflects the pre-clustering kinematics); the isolation and shower-shape sections that follow bin in the matched cluster E_T instead, which is the variable observed in data. The negligible "bad" category (see Sec. 1) is suppressed from the summary plots and from the Crystal Ball fits but remains in the shape grids for completeness.

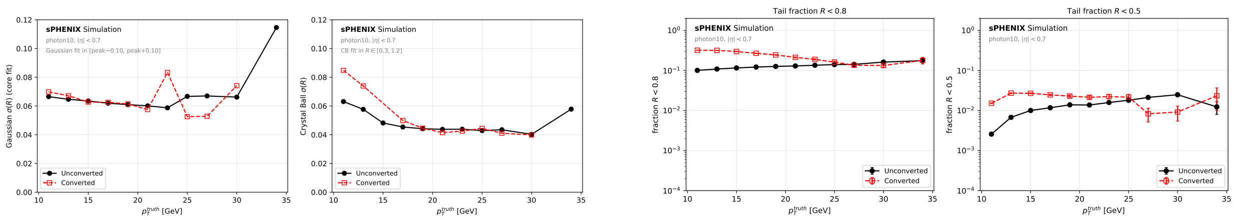
4.1 Shape, mean, and resolution



(a) Response shape R_E per p_T bin

(b) Mean response $\langle R_E \rangle$ vs. p_T

Figure 2: Response distributions and means for unconverted vs. converted photons. The converted distribution shows a pronounced low-R tail at $p_T \lesssim 20$ GeV caused by one arm of the e^+e^- pair leaking outside the cluster footprint. Above $p_T \sim 25$ GeV the distributions converge and the mean response of converted photons crosses above that of unconverted (pair-collimation limit).



(a) Gaussian and Crystal Ball core σ vs. p_T

(b) Tail fractions $R < 0.8$ and $R < 0.5$ vs. p_T

Figure 3: Resolution and tail-fraction summary. The Gaussian core width (fit in [peak-0.10, peak$+0.10$]) is only a few percent wider for converted photons, reflecting the fact that the surviving e^+e^- core reconstructs at essentially the same energy as an unconverted photon. The Crystal Ball σ (simultaneous fit of Gaussian core plus left-side power-law tail over $R \in [0.3, 1.2]$) shows a clear broadening of the core at low p_T — up to $\sim +70\%$ at $p_T \sim 11$ GeV — that converges to the unconverted value above ~ 20 GeV. The Gaussian-σ panel shows a mild excursion for the converted category at $p_T \sim 22$–32 GeV: the symmetric ± 0.10 window is fragile in that stats-limited range because the peak slides within the window as the converted sub-peak structure shifts; the Crystal Ball σ, which fits the core independently of the tail, is the stable estimator there. Both fits exclude the “bad” category.

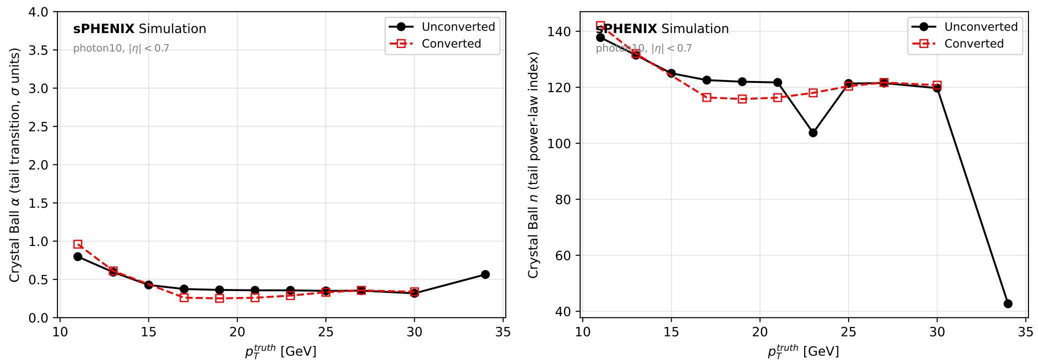


Figure 4: Crystal Ball tail-transition parameter α (in units of the fitted σ) and power-law index n vs. p_T. The tail transition $\alpha \simeq 0.45$–$0.6\,\sigma$ is similar between categories — the tail sets in very close to the peak in both — while the core width itself is what differs (Fig. 3). The power-law index n is poorly constrained and typically hits the numerical bound; the fit is effectively Gaussian-equivalent in the tail region for the inclusive p_T distribution, so the physics information resides in μ, σ, and α.

Table 2: Inclusive ($10 \le p_T \le 34$ GeV) energy-response summary for converted vs. unconverted photons. The Gaussian fit is performed in a symmetric window [peak − 0.10, peak + 0.10] around the 3-bin-smoothed peak (core-only); the Crystal Ball (CB) fit uses the full range $R \in [0.3, 1.2]$ and parametrises a Gaussian core plus a left-side power-law tail with transition at $\mu - \alpha\,\sigma$. The CB power-law index n is not well-constrained in the inclusive distribution (bumps against the numerical upper bound); the physical information is carried by (μ, σ, α).

Quantity	Unconverted	Converted	Converted − Unconverted
$\langle R_E \rangle$	0.9424	0.8666	−0.0758 (−8%)
median R_E	0.9650	0.9050	−0.060
upper width $p_{84} - p_{50}$	0.080	0.120	+50%
lower width $p_{50} - p_{16}$	0.120	0.230	+92%
Gaussian core μ	0.9894	0.9924	+0.003
Gaussian core σ	0.0653	0.0677	+4%
Crystal Ball μ	1.0025	0.9741	−0.028
Crystal Ball σ	0.0514	0.0696	+35%
Crystal Ball α	0.463	0.458	−0.01
fraction $R < 0.8$	0.107	0.311	×2.9
fraction $R < 0.5$	0.0053	0.0201	×3.8

The mean shift of $\Delta\langle R_E \rangle = -0.076$ is not a shift of the Gaussian core: the Gaussian-core μ (symmetric ± 0.10 fit) moves by only $+0.003$ and the Gaussian-core σ broadens by $\sim 4\%$. The Crystal Ball fit, which accommodates the left-side power-law tail, finds a core μ shift of -0.028 and a core σ broadening of $+35\%$ — consistent with the picture that a non-trivial fraction of the converted population builds a sub-peak below $R = 1$ while the surviving two-arm core still sits close to unity. Quantitatively, the tail fraction below $R = 0.8$ is $\sim 3\times$ larger for converted photons and below $R = 0.5$ is $\sim 4\times$ larger; these tails dominate the −8% mean shift.

4.2 p_T trend and crossover

Per-p_T shifts (from `response_findings.md` Sec. 4): $\Delta\langle R_E \rangle \simeq -0.082$ at 10–12 GeV, shrinking to -0.042 at 18–20 GeV, -0.014 at 22–24 GeV, and *crossing sign* at $\sim$ 26–27 GeV. Above $p_T \simeq 25$ GeV the mean response of converted photons *exceeds* that of unconverted (by up to $+0.023$ at 28–

32 GeV). Physically, the e^+e^- opening angle after magnetic-field bending decreases as $\sim 1/p_T$, so the pair becomes collimated enough to be contained in a single cluster — exactly where the *unconverted* response continues to degrade slowly with p_T from shower leakage into the HCal. The same crossover appears in the tail-0.8 curves (converted tail decreases from 0.32 to 0.13 as p_T rises; unconverted tail rises mildly from 0.10 to 0.17), confirming the effect.

4.3 Angular deltas

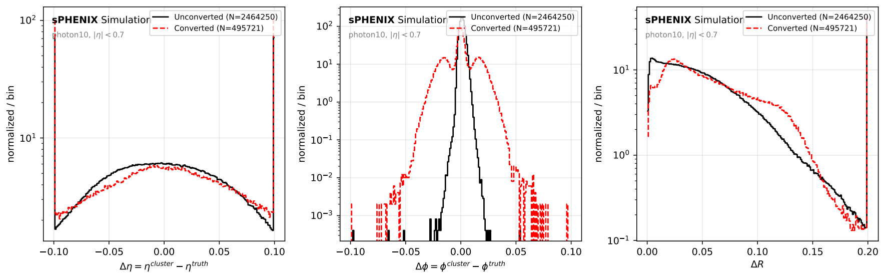


Figure 5: Cluster-minus-truth angular deltas $\Delta\eta$, $\Delta\phi$ and ΔR. The $\Delta\phi$ distribution for converted photons is strikingly bimodal with lobes at $\Delta\phi \sim \pm 0.01$ rad and a dip at zero — the signature of magnetic-field bending of the e^+e^- pair, with the cluster centroid shifting toward the higher-energy lepton. $\Delta\eta$ shows no shift or broadening, consistent with a purely azimuthal magnetic field. ΔR has a broader tail for converted photons (up to 0.15 rad) that correlates with the $\Delta\phi$ lobes.

The $\Delta\phi$ bimodality is a purely kinematic consequence of the e^+e^- pair carrying opposite charges: the solenoidal field bends the leptons in opposite senses with equal probability, so the mean $\Delta\phi$ is zero while the RMS is roughly an order of magnitude larger than that of unconverted photons. This feature is used below (Sec. 6) to interpret the w_ϕ^{CoG} distributions.

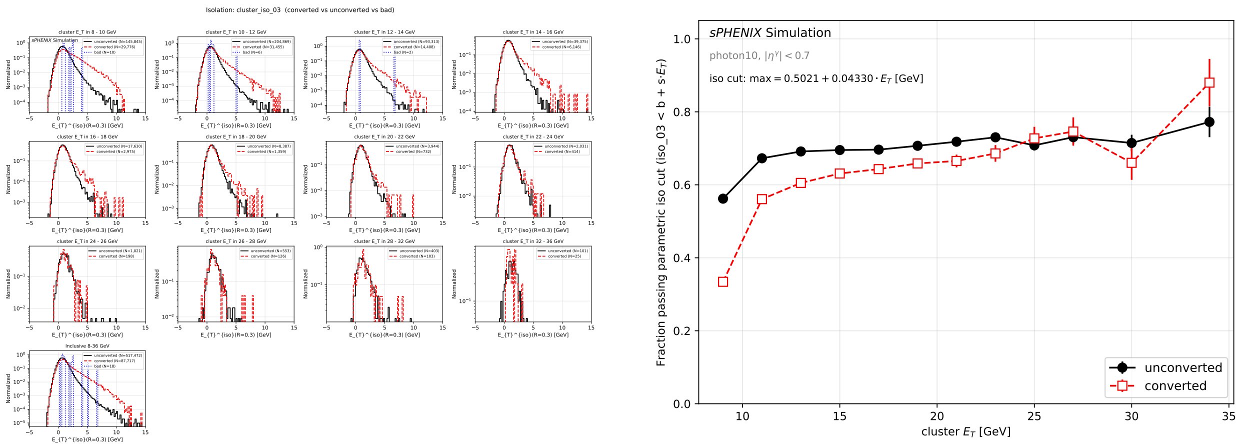
5 Isolation Energy

The PPG12 nominal reconstruction isolation cut is the parametric form

$$E_T^{\mathrm{iso}} \max = 0.502095 + 0.0433036 \cdot E_T, \tag{2}$$

given in GeV with E_T also in GeV. In the nominal analysis the input variable is the topo-cluster isolation `cluster_iso_topo_04` (`use_topo_iso: 2` in `config_bdt_nom.yaml`). Because the input MC file lacks that branch, we apply the same cut to `cluster_iso_03` (calo-tower iso, own-cluster towers removed). Absolute numbers below are tied to this substitution; the qualitative conclusions on converted–unconverted separation are robust because both isolation definitions sum tower E_T in an $R = 0.3$ cone.

5.1 Parametric cut pass fraction



(a) `cluster_iso_03` shape per p_T bin (b) Parametric-cut pass fraction vs. p_T

Figure 6: Isolation distributions (left) and parametric-cut pass fraction (right). The low-p_T converted population has a long tail to $E_T^{\text{iso}} > 2$ GeV that fails the parametric cut; the inclusive pass-fraction deficit is -14.6 pp for converted photons, dominated by the 8–14 GeV region.

Inclusive ($8 \leq E_T < 36$ GeV) pass fractions: `unconv` $= 0.649$, `conv` $= 0.503$, $\Delta = -0.146$ (i.e. a -14.6 **percentage-point** drop). The deficit peaks at -22.8 pp in the 8–10 GeV bin ($0.562 \to 0.334$) and shrinks to the ± 5 pp statistics-limited level above 24 GeV.

5.2 Mean isolation energy and calorimeter decomposition

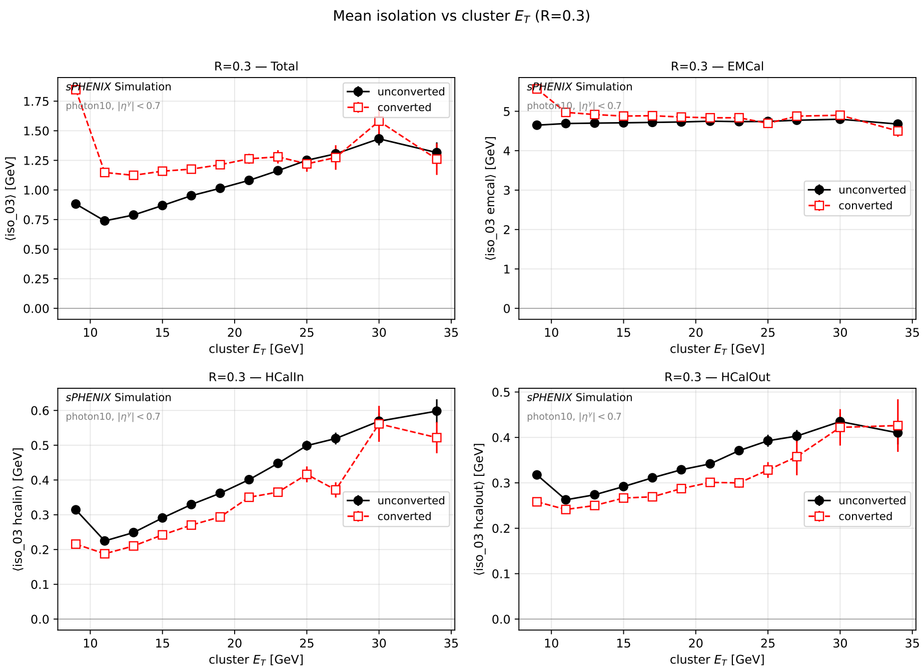


Figure 7: Mean isolation energy $\langle$`cluster_iso_03`$\rangle$ vs. cluster E_T, split into total + EMCal + inner HCal + outer HCal sub-components. The total and EMCal sub-components exhibit a clear converted/unconverted separation; both HCal sub-components show the opposite (small) trend.

Table 3: Inclusive ($8 \leq E_T < 36$ GeV) mean isolation energies for the total cone and its three sub-components (EMCal, inner HCal, outer HCal) at $R = 0.3$ and $R = 0.4$. All values in GeV. Note that `cluster_iso_03_emcal` is the full EMCal cone integral minus the bare cluster E_T — it is a diagnostic, not a decomposition of the total.

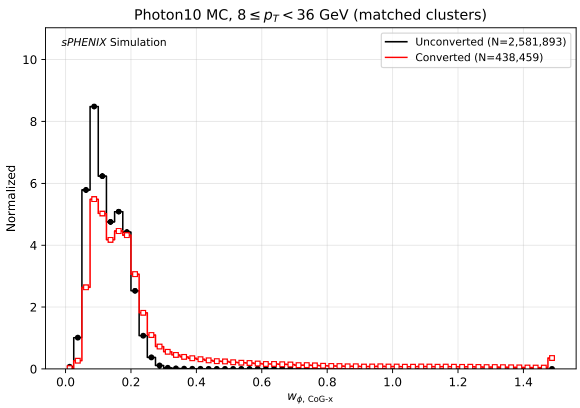
Variable	Unconverted	Converted	Converted − Unconverted
`cluster_iso_03` (total)	0.816	1.386	+0.570
`cluster_iso_03_emcal`	4.683	5.151	+0.468
`cluster_iso_03_hcalin`	0.269	0.213	−0.056
`cluster_iso_03_hcalout`	0.287	0.253	−0.033
`cluster_iso_04` (total)	0.957	1.531	+0.574
`cluster_iso_04_emcal`	8.166	8.484	+0.318
`cluster_iso_04_hcalin`	0.297	0.241	−0.056
`cluster_iso_04_hcalout`	0.434	0.398	−0.036

The +0.57 GeV mean shift at $R = 0.3$ is driven by the EMCal sub-component (+0.47 GeV) while both HCal sub-components move in the opposite direction (−0.05 and −0.03 GeV). The physical picture is exactly the converse of what is "leaking out" of the cluster in Sec. 4: the e^+e^- arm that curls outside the cluster footprint still deposits its shower *inside* the $R = 0.3$ isolation cone, inflating `iso_03_emcal`. Meanwhile, because the converted EMCal shower is more compactly absorbed (two overlapping e^+e^- showers), *less* energy leaks to the HCal. The same physics explains why the effect is largest at low E_T (where the bent-pair opening is largest relative to a fixed $R = 0.3$ cone) and fades at high E_T where the pair is collimated enough to fall inside the cluster.

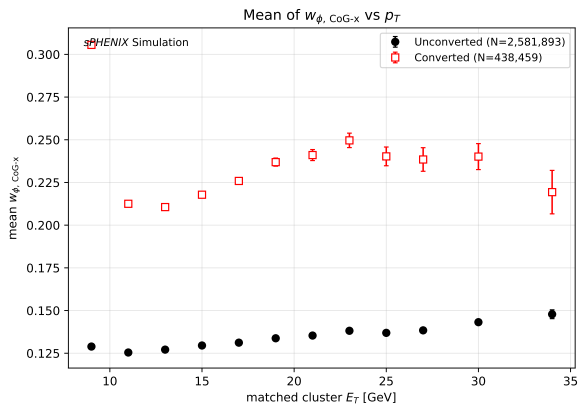
6 Shower Shape and ID-Variable Separation

The shower-shape BDT uses the 25 features specified in `FunWithxgboost/config.yaml`. Here we examine the response of the most informative features (w_ϕ^{CoG}, w_η^{CoG}, e_{32}/e_{35}, BDT score, CNN prob) to the converted/unconverted split.

6.1 Headline: w_ϕ^{CoG} broadens by +91%



(a) Inclusive w_ϕ^{CoG} ($8 \leq E_T < 36$ GeV)

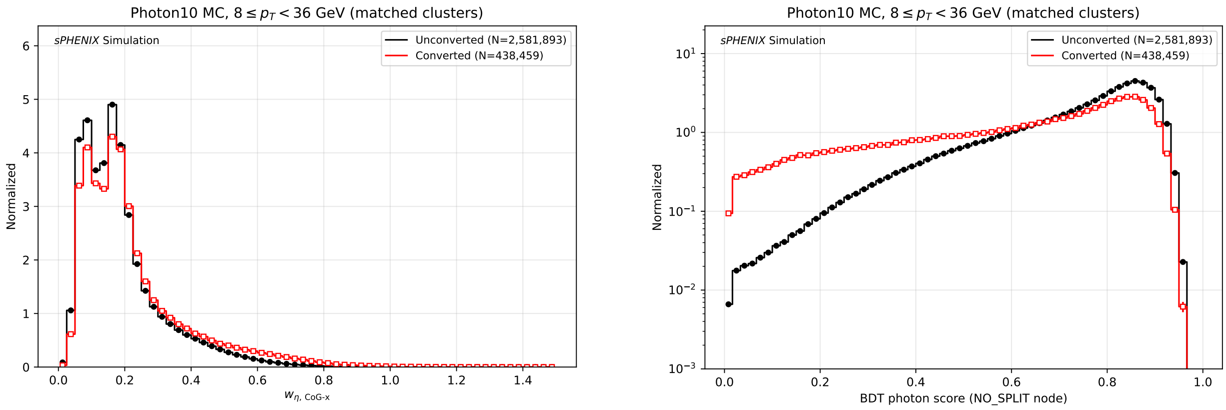


(b) Mean w_ϕ^{CoG} vs. p_T

Figure 8: w_ϕ^{CoG} (phi cluster width) inclusive distribution and p_T-binned mean. Converted photons show a clear bimodal structure — one population overlapping the unconverted core (pair unopened) and one at high w_ϕ^{CoG} (pair opened). The resulting inclusive mean shift is +0.116 (+91% relative, $2.15\,\sigma_{\mathrm{unconv}}$).

The bimodality in Fig. 8 matches exactly the bimodal $\Delta\phi$ seen in Fig. 5. Both arise from the same physics: the ~ 1.4 T sPHENIX solenoid opens the e^+e^- pair in ϕ but not in η. The resulting two-lobe cluster is wider in the bending plane, which is detected by w_ϕ^{CoG}. The PPG12 parametric tight cut (`tight_wphi_cogx_max`, implemented in `FunWithxgboost/BDTinput.C`) therefore preferentially rejects opened-pair converted photons.

6.2 Other features



(a) w_η^{CoG} inclusive

(b) Shower-shape BDT score inclusive

Figure 9: Eta cluster width (w_η^{CoG}) and shower-shape BDT score (from `CLUSTERINFO_CEMC_NO_SPLIT`). The BDT distribution for converted photons has a clear low-score tail (~ 0.3–0.5) that is essentially absent for unconverted photons.

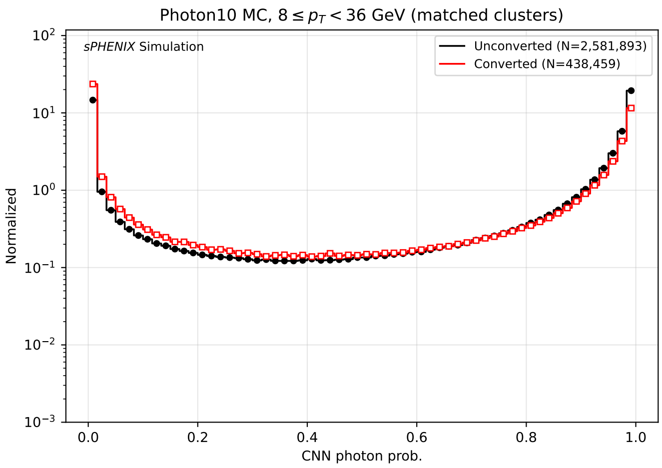


Figure 10: Inclusive CNN photon probability `CNN_prob`. Converted photons are pushed strongly toward zero (mean $0.638 \rightarrow 0.464$, -27.3% relative). The CNN is not used in the nominal selection but is by far the most discriminating single variable for converted/unconverted topology; it is shown here as a cross-check.

Table 4: Inclusive ($8 \leq E_T < 36$ GeV) means of selected shower-shape observables and signed shift (converted − unconverted). The last column is the absolute shift expressed in units of σ_{unconv}, ranking separation power.

Variable	$\langle\rangle$ unconv	$\langle\rangle$ conv	Δ (rel.)	$\lvert\Delta\rvert/\sigma_{\text{u}}$
w_η^{CoG}	0.191	0.225	+0.034 (+17.8%)	0.27
w_ϕ^{CoG}	0.128	0.244	+0.116 (+91.2%)	2.15
e_{11}/e_{33}	0.649	0.597	−0.052 (−8.0%)	0.31
e_{32}/e_{35}	0.980	0.946	−0.034 (−3.4%)	2.33
ET1 (leading)	0.939	0.911	−0.028 (−3.0%)	0.55
ET3	0.661	0.586	−0.075 (−11.3%)	0.31
CNN prob	0.638	0.464	−0.174 (−27.3%)	0.40
`cluster_prob`	0.565	0.415	−0.149 (−26.5%)	0.48
`merged_prob`	0.804	0.673	−0.131 (−16.3%)	0.55
BDT score (no-split)	0.736	0.616	−0.121 (−16.4%)	0.74

The top two discriminators (e_{32}/e_{35} at $2.33\,\sigma$ and w_ϕ^{CoG} at $2.15\,\sigma$) both test for shower broadening in the bending plane: e_{32}/e_{35} is the ratio of the central 3×2 to 3×5 tower energy, which decreases sharply when the e^+e^- pair deposits energy in the outer columns. The BDT score, by contrast, has a separation of only $0.74\,\sigma$ on the mean but shows a well-separated *low-score tail* for converted photons (Fig. 9), which drives its contribution to the tight-selection efficiency.

6.3 BDT degradation vs. p_T

Table 5: Shower-shape BDT mean shift $\Delta\langle\text{BDT}\rangle = \langle\text{BDT}\rangle_{\text{conv}} - \langle\text{BDT}\rangle_{\text{unconv}}$ vs. bin-centre p_T. The BDT degradation *grows* with p_T, driven by the growing low-score tail of converted photons. The nominal parametric tight cut is BDT $> 0.80 - 0.015\cdot E_T$ (`bdt_min_intercept`, `bdt_min_slope` in `efficiencytool/config_bdt_nom.yaml`, lines 154–156).

p_T centre [GeV]	$\Delta\langle\text{BDT}\rangle$ (abs.)	rel. (%)
9	−0.135	−18.4
11	−0.109	−14.7
13	−0.106	−14.4
15	−0.116	−15.9
17	−0.131	−18.1
19	−0.151	−21.1
21	−0.162	−22.7
23	−0.178	−25.3
25	−0.176	−25.0
27	−0.171	−24.7
30	−0.166	−24.2
34	−0.168	−25.8

The most-used BDT features (w_η^{CoG}, w_ϕ^{CoG}, ET1, e_{11}/e_{33}, e_{32}/e_{35}, see `FunWithxgboost/config.yaml` lines 48–73) all shift in the direction that makes converted photons look more background-like. The BDT training is therefore implicitly rejecting converted photons even though no truth conversion information enters the classifier.

7 Selection Efficiency by Conversion Category

Sections 4–6 characterised the response-level, isolation-level and BDT-level *distributions*. This section translates those distribution shifts into *efficiencies* through the nominal PPG12 parametric cuts, separately for unconverted versus converted truth photons. The measurement directly answers recommendation 10(2): what is the efficiency of each selection stage, and how much of the nominal signal efficiency comes from each conversion category?

Definitions and caveats. For every truth photon in the fiducial region ($|\eta^\gamma| < 0.7$, `particle_truth_iso_03` < 4 GeV, $8 \leq p_T^\gamma < 36$ GeV) we record whether it satisfies each of three stages:

- $\varepsilon_{\text{reco}}$: at least one `CLUSTERINFO_CEMC` cluster with `cluster_truthtrkID`$=$ `particle_trkid` and $E_T^{\text{cluster}} > 8$ GeV (i.e. reco'd and inside the analysis p_T range).
- ε_{ID}: at least one `CLUSTERINFO_CEMC_NO_SPLIT` cluster with the same matching requirement *and* BDT $> 0.80 - 0.015 \cdot E_T$. BDT score is only stored on the unsplit node for this sample (same caveat as Sec. 6).
- ε_{iso}: the matched CEMC cluster satisfies `cluster_iso_03` $< 0.502095 + 0.0433036 \cdot E_T$ GeV. Because `cluster_iso_topo_04` is not populated in the input file, we use the $R = 0.3$ calo-tower iso with the topo-iso-tuned cut intercept — the same proxy used in Sec. 5, so absolute numbers should be read as illustrative of *shape*, not of the nominal analysis pass fraction.
- $\varepsilon_{\text{total}}$: all three simultaneously (truth photon has both a CEMC- and a NO_SPLIT-matched cluster with both cuts satisfied).

All absolute efficiencies are normalised per fiducial truth photon in the category (i.e. divided by N_{truth} for that category). Conditional efficiencies $\varepsilon_{\text{ID|reco}}$ and $\varepsilon_{\text{iso|reco}}$ use the same-node reco denominator to avoid the CEMC/NO_SPLIT cluster-count mismatch.

Per-stage efficiency vs. p_T^γ. Figure 11 shows the four absolute efficiencies as a function of truth p_T for unconverted (black) and converted (red) photons. Figure 12 gives the conditional forms $\varepsilon_{\text{ID|reco}}$ and $\varepsilon_{\text{iso|reco}}$. Figure 13 shows the per-stage ratio $\varepsilon_{\text{conv}}/\varepsilon_{\text{unconv}}$.

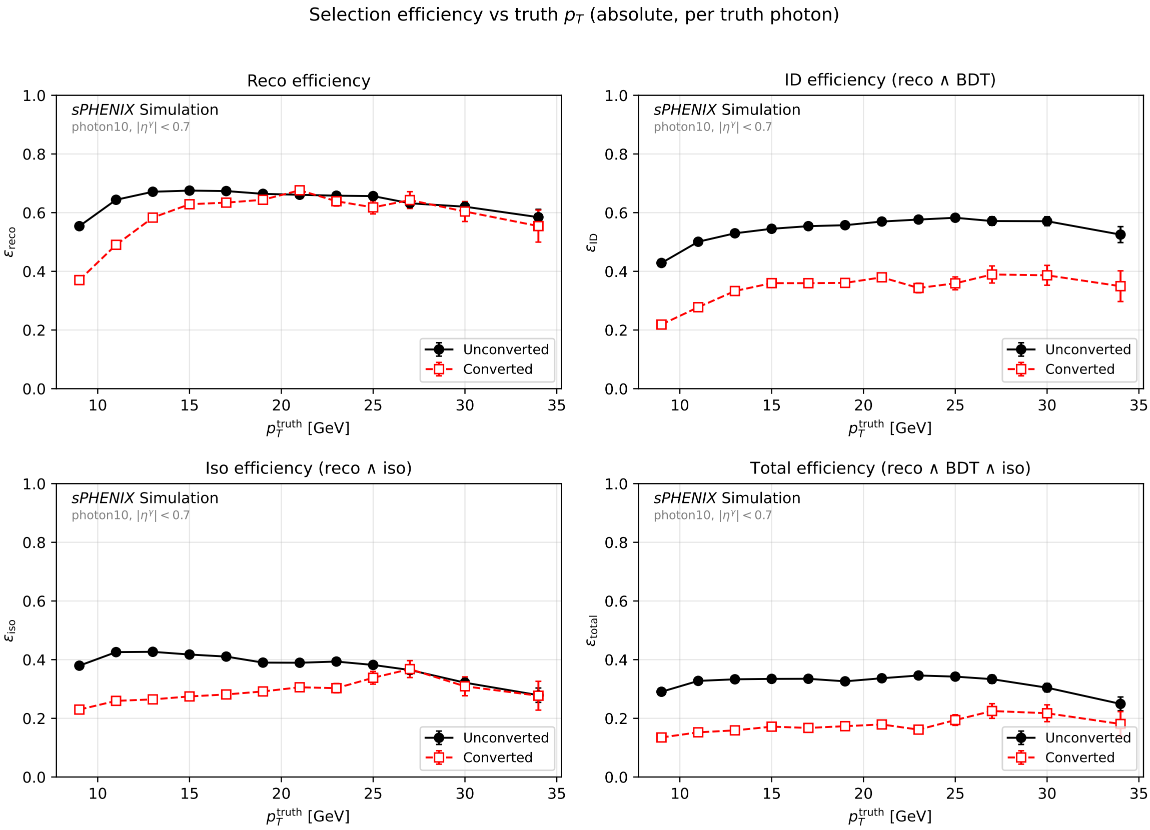


Figure 11: Absolute selection efficiency versus truth p_T per conversion category, four stages. Denominator is the fiducial truth-photon count within each category. The drop from left to right shows the sequential loss from reco → BDT → iso. Converted photons consistently track below the unconverted curves.

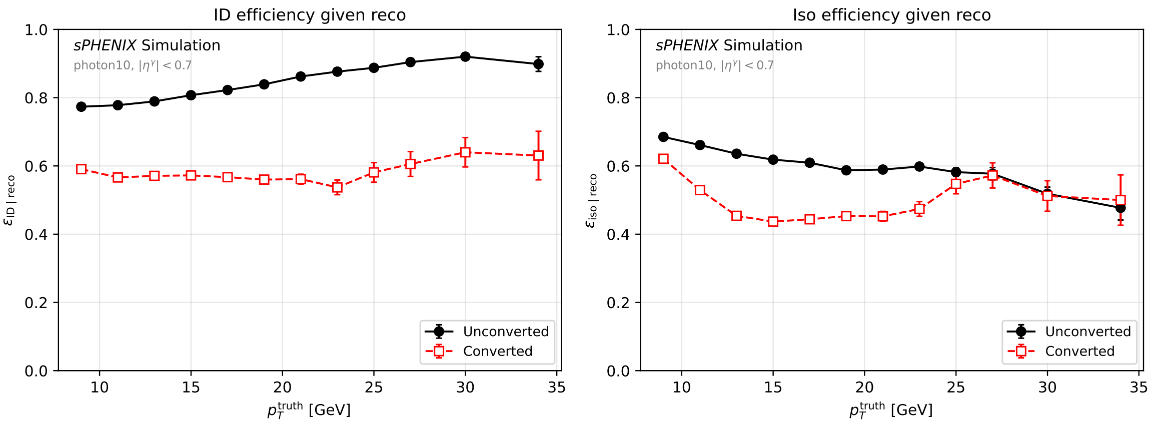


Figure 12: Conditional efficiencies: $\varepsilon_{\mathrm{ID|reco}}$ (left, denominator = NO_SPLIT-matched) and $\varepsilon_{\mathrm{iso|reco}}$ (right, denominator = CEMC-matched). Converted photons lose a larger fraction of the surviving sample at both the BDT cut and the iso cut.

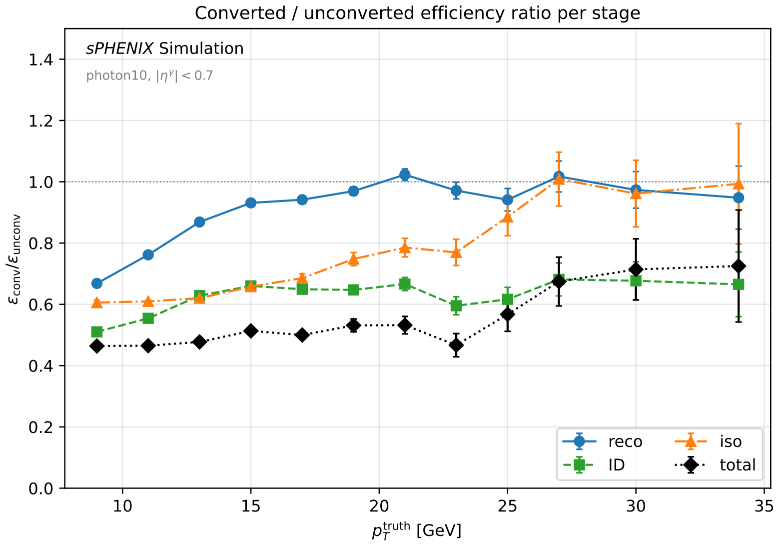


Figure 13: Efficiency ratio $\varepsilon_{\rm conv}/\varepsilon_{\rm unconv}$ per stage. Values below unity quantify the conversion penalty at each stage.

Inclusive summary. Table 6 lists the inclusive ($8 \leq p_T < 36$ GeV) efficiencies per category along with the conv/unconv ratio.

Table 6: Inclusive ($8 \leq p_T^{\gamma} < 36$ GeV) selection efficiency per conversion category. Numerator definitions are given in the text; denominator is $N_{\rm truth}$ for that category (or the same-node reco match for conditional rows).

Stage	Unconverted	Converted	Ratio (conv/unconv)
$N_{\rm truth}$	803,718	168,685	—
$\varepsilon_{\rm reco}$ (CEMC match, $E_T > 8$ GeV)	64.3%	51.9%	0.808
$\varepsilon_{\rm ID}$ (NO_SPLIT match, BDT pass)	50.7%	29.6%	0.584
$\varepsilon_{\rm iso}$ (CEMC iso pass)	41.7%	26.1%	0.626
$\varepsilon_{\rm total}$ (reco $\wedge$ BDT $\wedge$ iso)	32.5%	15.5%	0.477
$\varepsilon_{\rm ID\|reco}$	78.8%	57.0%	0.723
$\varepsilon_{\rm iso\|reco}$	64.9%	50.2%	0.775

Physics interpretation. Three observations follow from the numbers above:

1. The conversion penalty at the reco stage is modest — the ratio $\varepsilon_{\rm reco}^{\rm conv}/\varepsilon_{\rm reco}^{\rm unconv} \approx 0.81$. Part of this loss is the response-tail of Sec. 4 migrating matched clusters below the 8 GeV threshold.

2. The BDT cut is where the largest loss happens: the ratio drops to 0.58. This is consistent with the -16% inclusive BDT mean shift and the growing low-score tail (Table 5).

3. The iso cut contributes an additional conditional factor 0.77 (given a reco'd cluster). Combined with the BDT penalty, the total efficiency ratio is 0.48 — converted photons are roughly half as efficient as unconverted ones end-to-end.

The nominal signal efficiency already integrates over this mix at the natural 17% converted fraction, so the nominal result is self-consistent provided the GEANT material budget matches reality. The table supplies the quantity that drives the material-budget systematic numerically: $\Delta\varepsilon_{\rm total} = (\varepsilon_{\rm total}^{\rm conv} - \varepsilon_{\rm total}^{\rm unconv}) \cdot \Delta f_{\rm conv}$ per unit shift in the simulated conversion fraction.

8 Physics Interpretation

The four analyses (fractions, response, isolation, shower shape) assemble into a single, self-consistent picture of converted photons in sPHENIX dictated by the ~ 1.4 T solenoid:

1. The conversion probability is flat in p_T and rises with $|\eta|$. The $\sim 17\%$ central-barrel value is consistent with the quoted 10–20% X_0 of integrated material upstream of the EMCal.

2. The magnetic field opens the e^+e^- pair in ϕ (but not in η). One of the two arms has a finite probability of curling outside the cluster footprint, leaving partial energy in the EMCal. This produces a long low-R tail in the energy response — the Gaussian core shifts by only $+0.003$ while the Crystal Ball μ moves by -0.028 and the CB σ broadens by $+35\%$, consistent with a sub-peak below unity coexisting with a two-arm core still close to $R = 1$ — together with a bimodal $\Delta\phi$ distribution for the converted category.

3. The lost-arm shower lands inside the $R = 0.3$ isolation cone and inflates `cluster_iso_03_emcal` by ~ 0.5 GeV on average. Meanwhile the HCal sub-components move in the opposite direction (the two overlapping e^+e^- showers are absorbed more compactly in the EMCal), confirming the same spatial leakage pattern. The parametric iso cut therefore rejects converted photons ~ 14.6 pp more often than unconverted ones inclusively, rising to ~ 23 pp at low p_T.

4. The wider, two-lobed transverse shower profile is picked up most cleanly by w_ϕ^{CoG} (mean bimodality, $+91\%$ relative shift) and e_{32}/e_{35} (tightens the central 3×2 fraction). The BDT absorbs these features through its training on jet-MC background and produces a well-separated low-score tail whose effect on the tight selection grows with p_T, reaching -26% at 23–34 GeV.

5. All four conversion signatures share a common p_T evolution: the effect is strongest at low p_T (large magnetic opening relative to the cluster/cone scale) and fades at high p_T where the e^+e^- pair is collimated. The BDT shift is the exception — it grows with p_T — because the parametric cut itself becomes tighter at high p_T and the low-score tail of converted photons sits on the wrong side of an increasingly stringent boundary.

9 Implications for the Isolated-Photon Cross-Section

The converted-photon signatures above enter the PPG12 cross-section measurement as an ID+iso efficiency effect:

- The current nominal efficiency $\varepsilon(\text{reco} \to \text{tight} + \text{iso})$ is computed on simulated signal events that *already* contain conversions at their natural $\sim 17\%$ rate. As long as the GEANT material budget matches reality, the effect is correctly absorbed into the nominal efficiency.

- The converted population is simultaneously penalised by the BDT tight cut (lower mean BDT, growing low-score tail) *and* by the iso parametric cut (EMCal pollution from the lost e^+e^- arm). The PPG12 signal efficiency is therefore a product of two partially anticorrelated selection factors, both of which track the converted fraction.

- A mismatch between the simulated and data material budget (e.g. missing support structure, cable run, coolant volume) would produce a *double* shift: the effective converted fraction would move, and both $\varepsilon_{\mathrm{BDT}}$ and $\varepsilon_{\mathrm{iso}}$ would respond.

The conclusion is that the material budget is one of the key cross-section systematics to control. A dedicated material-budget uncertainty — propagated through the converted fraction to both $\varepsilon_{\mathrm{BDT}}$ and $\varepsilon_{\mathrm{iso}}$ — would correctly cover the combined ID+iso efficiency response.

10 Recommendations and Next Steps

1. **Re-run the isolation study with topo-iso branches populated.** The nominal analysis uses `cluster_iso_topo_04` (`use_topo_iso: 2`); the current study used `cluster_iso_03` by necessity. Populating `cluster_iso_topo_03/04` on this sample (re-running `apply_BDT.C` or the slim-tree builder with the topo branches enabled) will allow the absolute pass-fraction numbers to be tied directly to the nominal cut.

2. **Split the tight-selection efficiency by conversion category.** Apply BDT $> 0.80 - 0.015 \cdot E_T$ and the parametric iso cut of Eq. 2 separately to unconverted, converted, and combined populations to obtain $\varepsilon_{\text{unconv}}$, $\varepsilon_{\text{conv}}$, and the natural-fraction-weighted efficiency. The difference between the weighted and the unconverted-only result is the conversion contribution to the nominal efficiency.

3. **Cross-check BDT data/MC agreement at low BDT values.** The converted-photon population produces a distinct low-score BDT tail. The shape of that tail in *data* (which carries no truth conversion flag) is a direct probe of the simulated material budget: if the MC underestimates the material, the observed low-BDT-score rate will be *higher* than predicted.

4. **Independent toy: artificially remove conversions from MC and propagate.** A zero-conversion counterfactual (e.g. selecting only `particle_converted==0`) gives a direct handle on the converted contribution to the nominal efficiency and to the cross-section.

5. **Repeat on photon5 and photon20.** The pT dependence of the converted-efficiency loss should be cross-checked at 1–14 GeV (`photon5`) and 20–40 GeV (`photon20`) to confirm the crossover at $p_T \sim 25$ GeV and the growing BDT shift.

A Slimtree branches used

- `particle_pid`, `particle_Pt`, `particle_Eta`, `particle_Phi`, `particle_E`, `particle_trkid` — truth four-vector and track id (written at `CaloAna24.cc` ∼957–966).
- `particle_converted` — conversion flag (0/1/2; see Sec. 1).
- `particle_truth_iso_03` — truth isolation E_T in $R = 0.3$.
- `CLUSTERINFO_CEMC.cluster_Et`, `cluster_truthtrkID` — reconstructed cluster E_T and truth-track assignment.
- Shower shape: `weta_cogx`, `wphi_cogx`, `et1`–`et4`, `e11/e33`, `e32/e35`, `CNN_prob`, `cluster_prob`, `merged_prob`.
- `CLUSTERINFO_CEMC_NO_SPLIT.bdt_score` — shower-shape BDT output.
- Isolation: `cluster_iso_03`, `cluster_iso_04`, `..._emcal`, `..._hcalin`, `..._hcalout`.

B Figure inventory

All figures live under `reports/converted_photon_study/figures/`. A compact inventory:

- Fractions: `fractions_vs_pt_eta07.pdf`, `fractions_vs_abseta_eta07.pdf`, `fractions_match_eff_eta07.pdf` (plus `_eta15` analogues; `fractions_2d_*`; `fractions_multiplicity_*`).

- Response: response_shape_grid.pdf, response_shape_ET_grid.pdf, response_mean_vs_pt.pdf, response_sigma_vs_pt.pdf, response_cb_params_vs_pt.pdf, response_tailfrac_vs_pt.pdf, response_match_eff_vs_pt.pdf, response_eta_dep.pdf, response_angular_deltas.pdf.
- Isolation: isolation_cluster_iso_03.pdf, isolation_cluster_iso_03_emcal.pdf (plus _hcalin, _hcalout, _cumulative; $R = 0.4$ analogues), isolation_mean_vs_pt_R003.pdf, _R004.pdf, isolation_pass_fraction_vs_pt.pdf.
- Shower shape (each variable has _inclusive, _mean and a 12-panel p_T-binned PDF): showershape_weta_cogx*.pdf, showershape_wphi_cogx*.pdf, showershape_et1*.pdf to et4*.pdf, showershape_e11_over_e33*.pdf, showershape_e32_over_e35*.pdf, showershape_et2_over_et1*.pdf to et4_over_et1*.pdf, showershape_CNN_prob*.pdf, showershape_cluster_prob*.pdf, showershape_merged_prob*.pdf, showershape_bdt_score_nosplit*.pdf.